Project: Worked examples
Title: Rolling bearing rating life
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

## 1. Scope and method

Equivalent dynamic load of a rolling bearing under radial and axial load, its basic rating life $L_{10}$ in millions of revolutions and in hours at the running speed, and the life modified for reliability, against a required life. $P = X \cdot F_r + Y \cdot F_a$. $L_{10} = (C/P)^p$ million revolutions, p = 3 for ball and 10/3 for roller bearings. $L_{10}h = L_{10} \cdot 10^6$ revolutions at speed n. $L_{nm} = a_1 \cdot L_{10}$, $L_{nmh} = a_1 \cdot L_{10}h$. The equation is used for $P \le 0.5 \cdot C$.
Ref: ISO 281 §6 — Basic dynamic load rating and basic rating life
Ref: Shigley's Mechanical Engineering Design Ch. 11 — Rolling-contact bearings

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| type | Bearing type (selects the life exponent) | ball | | drawing: DRG-GB-210 rev A | any |
| C | Basic dynamic load rating | 35.1 | kN | drawing: DRG-GB-210 rev A | 0.1 … 100000 kN |
| F_r | Radial load | 4 | kN | spec: LOAD-SCHEDULE-01 | 0 … 100000 kN |
| F_a | Axial load | 0 | kN | spec: LOAD-SCHEDULE-01 | 0 … 100000 kN |
| X | Radial load factor (bearing catalogue) | 1 | | drawing: DRG-GB-210 rev A | 0 … 2 |
| Y | Axial load factor (bearing catalogue) | 0 | | drawing: DRG-GB-210 rev A | 0 … 5 |
| n | Speed | 1500 | rpm | spec: LOAD-SCHEDULE-01 | 1 … 100000 rpm |
| a_1 | Life modification factor for reliability | 1 | | spec: SPEC-MECH-03 design criteria | 0.05 … 1 |
| L_req | Required life (0 for no criterion) | 5000 | h | spec: SPEC-MECH-03 design criteria | 0 … 1000000 h |

## 3. Assumptions

A1. Steady load and speed; the bearing is correctly fitted, lubricated and free of contamination. (Basic rating life conditions.)
A2. X and Y are the catalogue factors for this bearing and load ratio. (Supplied by the user.)

## 4. Working

`[10] p = 3` ISO 281 §6
`= 3`
Life exponent (ball bearing)

[11] P = X * F_r + Y * F_a ISO 281 §6
= 1 × 4 kN + 0 × 0 kN
= 4 kN
Equivalent dynamic load
[12] k_P = P / C
= 4 kN / 35.1 kN
= 0.11396
Load ratio P/C
[13] L_10 = (C / P)^p ISO 281 §6
= (35.1 kN / 4 kN)^3
= 675.68
Basic rating life (millions of revolutions)
[14] L_10h = L_10 * 1e6 * 2 * pi / n
= 675.68 × 1e6 × 2 × π / 1500 rpm
= 7507.6 h
Basic rating life in hours ($10^6$ revolutions of 2π rad at speed n)
[15] L_nm = a_1 * L_10
= 1 × 675.68
= 675.68
Modified rating life (millions of revolutions)
[16] L_nmh = a_1 * L_10h
= 1 × 7507.6 h
= 7507.6 h
Modified rating life in hours
[17] u_life = L_req / L_nmh
= 5000 h / 7507.6 h
= 0.666
Required over modified life

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| P | 4 kN | Equivalent dynamic load |
| k_P | 0.11396 | Load ratio |
| L_10 | 675.68 | $L_{10}$ (millions of revolutions) |
| L_10h | 7507.6 h | $L_{10}$ in hours |
| L_nm | 675.68 | Modified life (millions of revolutions) |
| L_nmh | 7507.6 h | Modified life in hours |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Modified life against required life | u_life = 0.666 | u_life ≤ 1 | 0.666 | ISO 281 §6 |

## 7. Applicability

No warnings.

## 8. Verification

Template mech.bearing-life@1.0.0 Content hash 01f73131b0af3ef6 Engine calc-engine/1.0.0
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